








Statistics of Financial Markets

Chapter 2: Classical distributions and stylised facts



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **BET**: Bucharest Exchange Trading (index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CDF**: Cumulative Distribution Function
- **CLT**: Central Limit Theorem
- **EGARCH**: Exponential GARCH
- **ES**: Expected Shortfall
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **JB**: Jarque–Bera (test)
- **LB**: Ljung–Box (test)
- **MLE**: Maximum Likelihood Estimation
- **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- **OUG**: Ordonanță de Urgență a Guvernului (government emergency ordinance)
- **QQ**: Quantile–Quantile (plot)
- **SFE**: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)

Today's Question, Route and Learning Outcomes

- **Question**: is the Normal distribution a good model for daily returns, and what do real returns look like?
  - the answer decides how we measure risk, price options and test hypotheses
- **Route and outcomes**: after this chapter you can
  - compute probabilities and quantiles of the Normal and lognormal distributions
  - state the central limit theorem (CLT) and say when the Normal approximation works
  - estimate skewness and excess kurtosis and test normality with the Jarque–Bera test
  - read a QQ plot and fit a Student-t distribution by maximum likelihood
  - check six stylised facts of returns Cont (2001) and explain what each means for a model
- Real data, 2010–2026: S&P 500, DAX, BET, Bitcoin and four BVB stocks (SNP, BRD, SNN, SNG)

Reading and Tools

- Textbook: Franke, Härdle & Hafner (2019), *Statistics of Financial Markets*, 5th ed., Sec. 3.3 (Normal and lognormal) and Sec. 11.2 (stylised facts)
  - exercises with solutions: Borak, Härdle & López-Cabrera (2013)
- Complementary: Tsay (2010), Sec. 1.2 (distributional properties of returns); Cont (2001)
- Python Quantlets of this chapter: Quantlets/Ch_02
  - ported from the SFE Quantlets of the textbook (SFEDaxReturnDistribution, SFElognormal, SFEclt, SFENormalApprox1–4)
  - each chart in these slides links to the Quantlet that draws it
- Lecture notebook: open in Google Colab
- Video course: Tukey's g- and h- transformations

A Day That Should Not Happen

- 16 March 2020: the S&P 500 fell by -12.8% (log return) in one day
- Daily mean and standard deviation 2010–2026: 0.045% and 1.084%, so the fall was $(-12.77 - 0.045)/1.084 = -11.8$ standard deviations
  - under the Normal distribution, the probability of such a day is about 2×10^{-32}
  - that is, less than once in 10^{29} years of trading
- Yet such days happen: 19 October 1987, October 2008, March 2020, April 2025
- Conclusion: the model, not the market, is wrong

Question for the Room: How Often Is a 4-Sigma Day?

- A k-**sigma day**: a daily return more than k standard deviations away from the mean
- **What do you think?** If daily returns were Normal, how often would a 4-sigma day occur?
- **What do you think?** How often does it occur in the S&P 500 since 2010?
- **Answer**
  - Normal distribution: once every 15 787 trading days, about once in 63 years
  - S&P 500: 28 days in 4 201 days, once every 150 days, about 1.7 per year

What a Distribution Is Used For

- **Risk**: the probability of a large loss
  - VaR 1% (Value at Risk): the loss exceeded with probability 1%; a quantile of the distribution (Chapter 10)
- **Pricing**: the Black–Scholes formula assumes lognormal prices
  - heavier tails make deep out-of-the-money options more valuable
- **Inference**: *t*-tests and confidence intervals assume (approximate) normality
  - heavy tails make standard errors unreliable in small samples
- A wrong distribution gives wrong numbers, even with perfect data

The Benchmark Model of This Chapter

- Price P_t, log return $r_t = \ln(P_t/P_{t-1})$, so $P_t = P_{t-1}e^{r_t} = P_0 e^{r_1+\cdots+r_t}$
  - notation: **i.i.d.** = independent and identically distributed
- **Benchmark**: r_t i.i.d. $N(\mu, \sigma^2)$
  - $N(\mu, \sigma^2)$: the Normal distribution with mean μ and variance σ^2 (daily mean and variance of the log returns)
  - prices follow a geometric random walk; P_t is lognormal Osborne (1959)
  - the first model of this kind, for arithmetic price changes: Bachelier (1900)
- This chapter tests each assumption of the benchmark in turn: normality, identical distribution and independence

Carl Friedrich Gauss and the Normal Distribution

- De Moivre (1733) found the bell curve as the limit of binomial probabilities
- Gauss (1809) used it for the errors of astronomical observations and derived least squares from it
- Laplace (1810) proved a first version of the central limit theorem
- Hence its other name: the **Gaussian** distribution
- In finance it became the default model of returns, because sums of Normal variables stay Normal



Carl Friedrich Gauss (1777–1855)

Portrait: Christian Albrecht Jensen (1840); public domain; Wikimedia Commons

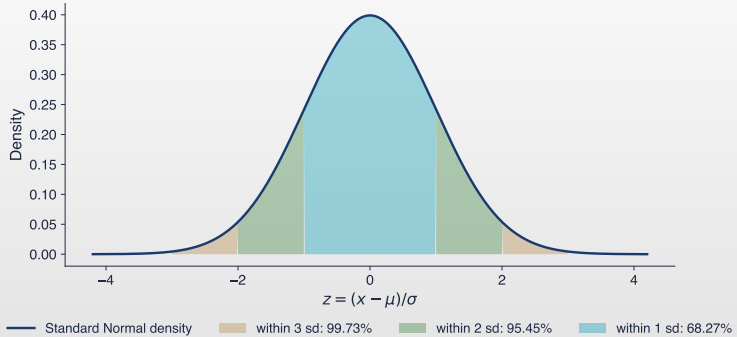
The Normal Distribution: Definition

- $X \sim N(\mu, \sigma^2)$ has the **PDF** (probability density function)
  - $f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} \exp\Big(-\dfrac{(x-\mu)^2}{2\sigma^2}\Big), \quad x \in \mathbb{R}$
  - two parameters: the mean μ (location) and the variance σ^2 (spread)
- **Standard Normal distribution**: $Z = (X - \mu)/\sigma \sim N(0, 1)$
  - its **CDF** (cumulative distribution function) is $\Phi(z) = P(Z \leq z)$, with no closed form; tables or software
- Probabilities of X come from Z: $P(X \leq x) = \Phi\big((x - \mu)/\sigma\big)$
- Quantile of order α: $x_\alpha = \mu + \sigma z_\alpha$, the value with $P(X \leq x_\alpha) = \alpha$
  - $z_\alpha = \Phi^{-1}(\alpha)$: the quantile of $N(0, 1)$; $z_{0.01} = -2.33$, $z_{0.05} = -1.64$

Properties Used in Finance

- Symmetric around μ: mean = median = mode
- Skewness 0 and kurtosis 3 (defined in Section 5)
- **Closed under addition**: if $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ are independent, then $X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
  - h-day log return under the benchmark: $r_t(h) \sim N(h\mu, h\sigma^2)$
  - volatility grows with $\sqrt{h}$: the square-root-of-time rule used in Chapter 1
- Linear combinations of jointly Normal returns are Normal, so portfolio returns stay Normal
- Fully described by two numbers: once μ and σ are known, every probability is known

The 68–95–99.7 Rule



- Within 1, 2 and 3 standard deviations: 68.27%, 95.45% and 99.73% of the probability
- Outside 3 standard deviations: one day in 370; the tails of the Normal distribution fall off like $e^{-z^2/2}$, very fast

SFM_ch2_normal_lognormal

How Rare Are Large Moves under the Normal Distribution?

k	$P(\lvert Z\rvert > k)$	once every ... trading days	once every ... years
1	0.3173	3	0.013
2	0.0455	22	0.087
3	2.7×10^{-3}	370	1.5
4	6.3×10^{-5}	15 787	63
5	5.7×10^{-7}	1 744 278	6 922
6	2.0×10^{-9}	506 797 346	2 011 101

- $P(|Z| > k)$: the probability that a standard Normal variable lies more than k standard deviations from 0
- Years computed with 252 trading days a year
- Each extra standard deviation makes a move 7 to 291 times rarer
- A 6-sigma day should not happen in the whole history of stock markets

SFM_ch2_normal_lognormal

Worked Example: Large Losses of the S&P 500

- S&P 500, 2010–2026: $n = 4\,201$ daily log returns, mean 0.045%, standard deviation 1.08%
- A loss larger than 3%: $z = (-3 - 0.045)/1.08 = -2.81$
  - Normal distribution: $\Phi(-2.81) = 0.25\%$, so 10.5 days expected
  - observed: 49 days
- A loss larger than 5%: $z = -4.65$
  - Normal distribution: 6.9×10^{-3} days expected; observed: 8 days
- BET: 13 days below -5% against 5.1×10^{-3} expected
- The Normal distribution underestimates the frequency of large losses by orders of magnitude

Recap: The Normal Distribution

- Two parameters, symmetric, closed under addition, volatility scales with $\sqrt{h}$
- Its tails are thin: a 4-sigma day once every 63 years
- Real returns have many more large moves than it allows

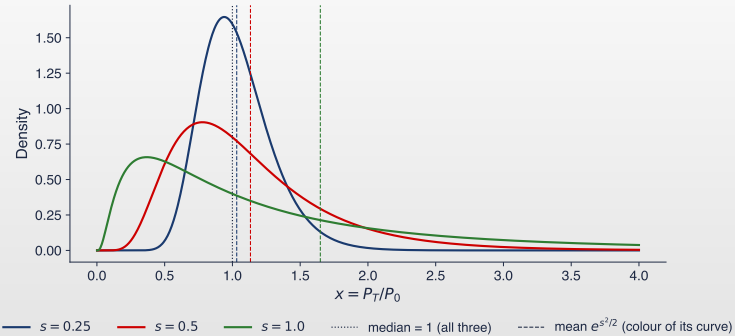
From Normal Log Returns to Lognormal Prices

- A positive variable Y is **lognormal**, $Y \sim LN(m, s^2)$, if $\ln Y \sim N(m, s^2)$
  - m, s^2: the mean and the variance of $\ln Y$, not of Y
- Density: $f(y) = \dfrac{1}{ys\sqrt{2\pi}} \exp\Big(-\dfrac{(\ln y - m)^2}{2s^2}\Big), \quad y > 0$
- Under the benchmark, $\ln(P_T/P_0) = r_1 + \cdots + r_T \sim N(T\mu, T\sigma^2)$
  - T: the horizon in days; so $P_T/P_0 \sim LN(T\mu, T\sigma^2)$ Osborne (1959)
- Why lognormal and not Normal prices?
  - prices cannot be negative; a Normal price could be
  - the simple return $R = e^r - 1 > -1$: you cannot lose more than you invested
- The Black–Scholes model is built on it

Mean, Median and Mode

- If $Y \sim LN(m, s^2)$:
- median e^m; mode e^{m-s^2}; mean $E[Y] = e^{m+s^2/2}$
  - mode: the most likely value; median: the value exceeded with probability 1/2
  - always mode $<$ median $<$ mean: a long right tail (positive skewness)
- Variance: $\mathrm{Var}(Y) = (e^{s^2} - 1)\, e^{2m+s^2}$
- The mean comes from $E[e^X] = e^{m+s^2/2}$ for $X \sim N(m, s^2)$ (Seminar 2 derives it)
  - the extra $s^2/2$ is the volatility drag of Chapter 1, seen from the other side
- Which one matters? The median is the outcome of a typical investor; the mean is pulled up by a few very good paths

Lognormal Densities (SFElognormal)



- $m = 0$: the median is 1 for all three curves; the dashed lines mark the means
- $s = 0.25$: mean 1.03, mode 0.94, almost symmetric; $s = 1$: mean 1.65, mode 0.37, skewness 6.18
- Long horizons or high volatility: the gap between mean and median widens

SFM_ch2_normal_lognormal

Worked Example: 100 Lei in the BET

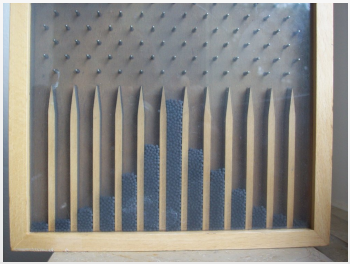
- BET, 2010–2026: annual mean log return $\mu = 11.6\%$ and volatility $\sigma = 16.9\%$ (daily moments $\times$ 250.6 and $\sqrt{250.6}$)
- After 1 year, $P_{\mathbf{1}}/P_{\mathbf{0}} \sim LN(\mu, \sigma^{\mathbf{2}})$
  - median $100e^{\mu} = 112.3$; mean $100e^{\mu+\sigma^{\mathbf{2}}/\mathbf{2}} = 113.9$
  - probability of a loss $\Phi(-\mu/\sigma) = 24.8\%$; 5% quantile $100e^{\mu-\mathbf{1.645}\sigma} = 84.9$
  - $\Phi(-\mu/\sigma) = P(r < 0)$: the probability of a negative log return; $-1.645 = z_{\mathbf{0.05}}$
- After 10 years, $P_{\mathbf{10}}/P_{\mathbf{0}} \sim LN(10\mu, 10\sigma^{\mathbf{2}})$
  - median 317.7; mean 366.8; probability of a loss 1.5%; 5% quantile 131.6
- The probability of a loss falls with the horizon; after 10 years even the 5% quantile is above 100, but only if the historical mean persists
- Caveat: these numbers inherit every weakness of the Normal model of log returns

Recap: The Lognormal Distribution

- Normal log returns $\Rightarrow$ lognormal prices: positive, right-skewed
- Median e^m below mean $e^{m+s^2/2}$: the gap is the volatility drag
- Report the median and a low quantile, not only the mean

The Galton Board

- Francis Galton (1889): balls fall through rows of pins and bounce left or right with probability 1/2
- The final position of a ball is a **sum** of many small independent shocks
- The heaps form a bell curve: binomial probabilities close to the Normal density
- The same argument was used for returns: a daily return is the sum of many small price changes during the day



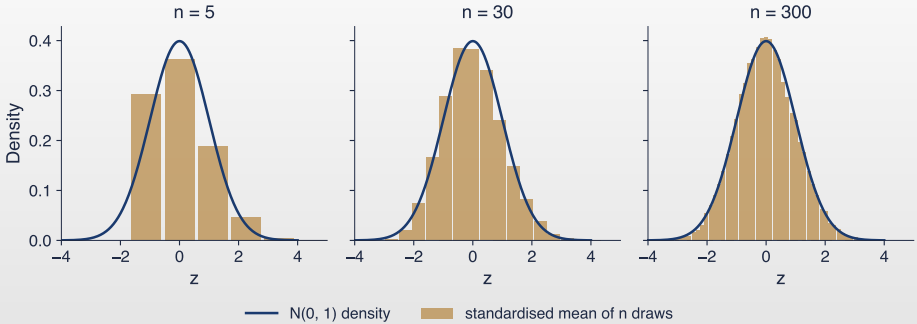
A Galton board after the balls have fallen

Photo: Klaus-Dieter Keller (2010); public domain; Wikimedia Commons

The Central Limit Theorem

- **CLT** (Lindeberg–Lévy): $X_1, \ldots, X_n$ i.i.d. with mean μ and finite variance σ^2
  - $\dfrac{\sqrt{n}\,(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1)$ as $n \to \infty$
  - $\bar{X}_n$: the mean of the n variables; $\xrightarrow{d}$: convergence in distribution, the CDF of the left side approaches Φ
  - equivalently, the sum $X_1 + \cdots + X_n$ is approximately $N(n\mu, n\sigma^2)$
- Three conditions
  - independence; identical distribution; finite variance
- How fast? The error falls like $1/\sqrt{n}$ and grows with the skewness of X (Berry–Esseen bound)
- It explains why the Normal distribution is everywhere, and why sample means have Normal confidence intervals

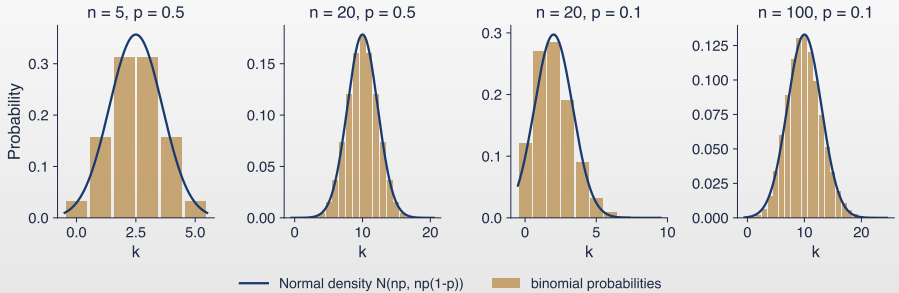
The CLT at Work (SFEclt)



- Standardised mean of n Bernoulli(0.2) draws (a skewed variable), 20 000 simulations
- Share above 1.96: 5.8% for $n = 5$, 2.7% for $n = 30$, 2.8% for $n = 300$; Normal distribution: 2.5%
- Skewness of the mean, $(1 - 2p)/\sqrt{np(1-p)}$ with $p = 0.2$: 0.67, 0.27, 0.09; it falls like $1/\sqrt{n}$

SFM_ch2_clt_normal_approx

The Normal Approximation of the Binomial (SFENormalApprox)



- $B(n, p)$: the number of successes in n independent trials with success probability p; $B(n, p) \approx N\big(np, np(1-p)\big)$
- Continuity correction: $P(X \le k) \approx \Phi\big((k + 0.5 - np)/\sqrt{np(1-p)}\big)$; the $+0.5$ spreads each integer k over $[k - 0.5, k + 0.5]$
- Largest error of the CDF: 0.001 for $n = 20, p = 0.5$; 0.037 for $n = 20, p = 0.1$; 0.017 for $n = 100, p = 0.1$
- Rule of thumb: the approximation is good when $np(1-p)$ is large (at least about 9)

SFM_ch2_clt_normal_approx

Question for the Room: Does the CLT Make Daily Returns Normal?

- A daily log return is the sum of thousands of intraday log returns
- **What do you think?** Should daily returns then be close to the Normal distribution?
- **Answer**: not necessarily; each condition of the CLT can fail
  - dependence: calm and turbulent days come in clusters (Section 8)
  - changing distribution: volatility differs from day to day
  - infinite variance: Mandelbrot (1963) proposed it for cotton prices (Chapter 3, α-stable distributions)
- The data decide: the next sections measure how far returns are from the Normal distribution

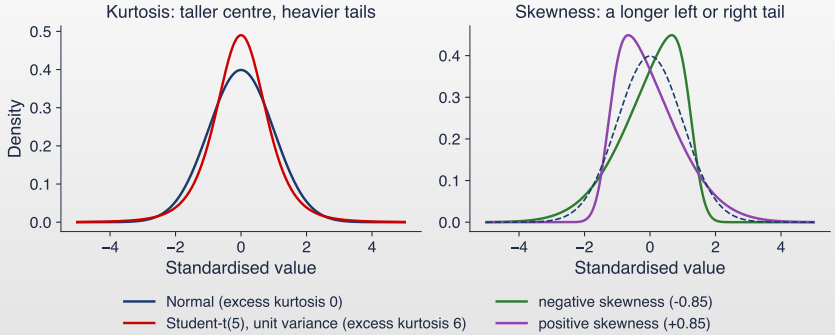
Recap: The Central Limit Theorem

- Sums of many i.i.d. terms with finite variance are approximately Normal
- Convergence is slower for skewed variables; $np(1-p) \geq 9$ for the binomial
- Returns may violate independence, identical distribution or finite variance

Four Moments

- Mean $\mu = E[X]$; variance $\sigma^2 = E[(X-\mu)^2]$; $E[\cdot]$ is the expected value
- **Skewness**: $\gamma_1 = E[(X-\mu)^3]/\sigma^3$
  - the third power keeps the sign of each deviation; $\gamma_1 < 0$: a longer left tail, large losses more frequent than large gains
- **Kurtosis**: $\gamma_2 = E[(X-\mu)^4]/\sigma^4$; Normal distribution: $\gamma_2 = 3$
  - **excess kurtosis** $\gamma_2 - 3$: positive means heavy tails and a tall, narrow centre (**leptokurtic**)
- Both are free of units: they describe the shape, not the location or the scale
- Fourth powers make the kurtosis very sensitive to a few extreme days

What Skewness and Kurtosis Look Like



- Left: same mean and variance; the Student-$t(5)$ curve (Section 7) has excess kurtosis 6: more mass in the centre and in the tails, less in the shoulders
- Right: skew-Normal densities with skewness ± 0.85; the dashed curve is the Normal distribution

SFM_ch2_moments_jarque_bera

Sample Moments

- Central sample moments: $m_k = \frac{1}{n}\sum_{t=1}^{n}(r_t - \bar{r})^k$, the average k-th power of the deviations from the mean (m_2: variance)
- Sample skewness $S = m_3/m_2^{3/2}$; sample excess kurtosis $K = m_4/m_2^2 - 3$
- If the data are i.i.d. Normal: $S \approx N(0, 6/n)$ and $K \approx N(0, 24/n)$
  - with $n = 4\,201$: standard errors $\sqrt{6/n} = 0.038$ and $\sqrt{24/n} = 0.076$
- These standard errors hold only under normality; for heavy-tailed data they are far too small (Seminar 2, B1)

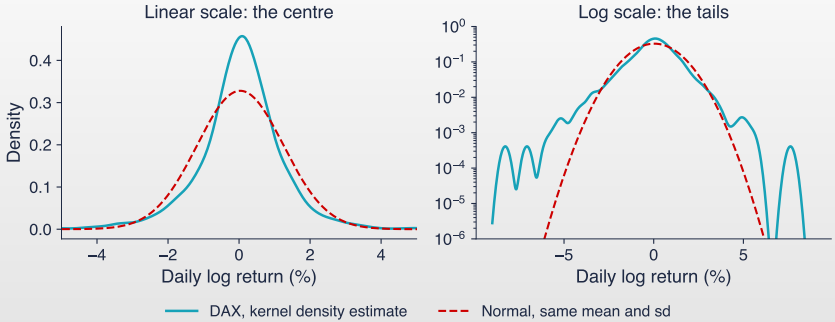
Daily Log Returns: Moments, 2010–2026

Series	from	n	Mean %	S.d. %	Skew.	Exc. kurt.	Min %
S&P 500	2010	4 201	0.045	1.08	−0.61	13.5	−12.8
DAX	2010	4 244	0.034	1.22	−0.45	7.4	−13.1
BET	2010	4 186	0.046	1.07	−0.98	17.9	−11.9
Bitcoin	2014	4 384	0.118	3.49	−0.70	12.0	−46.5
OMV Petrom (SNP)	2010	3 815	0.076	1.62	−0.54	8.9	−15.8
BRD	2010	3 771	0.044	1.66	−0.38	7.3	−18.2
Nuclearelectrica (SNN)	2013	2 911	0.099	1.52	0.27	5.1	−9.4
Romgaz (SNG)	2013	2 909	0.090	1.40	−0.34	5.3	−11.0

- Every series has excess kurtosis between 5.1 (SNN) and 17.9 (BET): far from 0
- Every series except SNN has negative skewness
- Worst days: S&P 500 on 16 March 2020, BET on 19 December 2018 (OUG 114, a tax on bank assets), Bitcoin on 12 March 2020

SFM_ch2_moments_jarque_bera

DAX Returns vs the Normal Density (SFEDaxReturnDistribution)



- Kernel density estimate (a smoothed histogram) against the Normal density with the same mean and standard deviation
- Centre: peak 0.46 vs 0.33; 76.8% of days within one standard deviation, vs 68.3% for the Normal distribution
- Tails (log scale): at -6% the estimated density is 439 times the Normal one

SFM_ch2_moments_jarque_bera

Check the Data Before Measuring Tails

- Banca Transilvania (TLV), 27–31 May 2016: the close falls from 2.83 to 2.09 lei, the ex-date of a bonus share issue
- The adjusted close in our data: 3.6014, then 3.0995, then 3.8060
  - log returns -15.0% and $+20.5\%$ on two consecutive days: the adjustment was applied one day late
- Excess kurtosis of TLV, 2010–2026: 17.5 with these two days, 13.5 without them
- Two wrong days out of thousands change the kurtosis by a third: this is why TLV is not in the tables of this chapter
- Rule from Chapter 1: list the largest moves and check each one, especially pairs of opposite jumps

Recap: Moments

- Skewness measures asymmetry, excess kurtosis measures tail weight relative to the Normal distribution
- Daily returns: mostly negative skewness and excess kurtosis between 5 and 18
- Kurtosis is driven by a few days: check them before trusting it

The Jarque–Bera Test

- H_0: the data are Normal, so skewness $= 0$ and excess kurtosis $= 0$
- **JB** statistic Jarque & Bera (1980), Jarque & Bera (1987): $\text{JB} = \frac{n}{6}\left(S^2 + \frac{K^2}{4}\right)$
  - the sum of two squared standardised statistics: $(S/\sqrt{6/n})^2 + (K/\sqrt{24/n})^2$
- Under H_0 and for large n: $\text{JB} \sim \chi^2(2)$; reject at 5% if $\text{JB} > 5.99$
  - $\chi^2(2)$: the chi-squared distribution with 2 degrees of freedom, the distribution of a sum of two squared independent $N(0,1)$ variables; 5.99 is its 95% quantile
  - $\text{JB} = 0$ for a perfect Normal shape; large values mean skewness or excess kurtosis far from 0
- Simple, uses only two moments, and is the standard test reported in empirical finance

Worked Example: Jarque–Bera for the S&P 500

- $n = 4\,201$, $S = -0.606$, $K = 13.49$
- $\text{JB} = 700.17 \times (0.3673 + 45.497) = 32\,113$
- Critical value 5.99: normality is rejected; the p-value is zero to any number of decimals
- The kurtosis term dominates: almost all of JB comes from $K^2/4$
- JB for the other series: all rejected at any usual level
  - indices and Bitcoin: BET 56 762, DAX 9 909, Bitcoin 26 482
  - BVB stocks: SNP 12 832, BRD 8 370, SNN 3 174, SNG 3 415

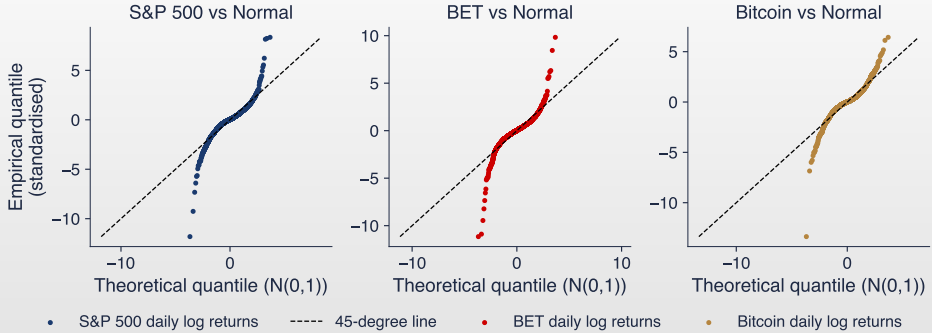
What Jarque–Bera Does Not Tell You

- With thousands of observations it rejects even tiny departures
  - report the size of S and K, not only the p-value
- It assumes i.i.d. data; with volatility clustering its standard errors are too small
  - S&P 500: Normal-theory 95% interval for K: $[13.34, 13.64]$; bootstrap interval: $[6.7, 20.6]$ (Seminar 2, B1)
- A rejection says the data are not Normal; it does not say which distribution they follow
- Other tests (Kolmogorov–Smirnov, Anderson–Darling) compare the whole CDF: Chapter 6

QQ Plots

- A **QQ plot** (quantile–quantile plot) compares the empirical quantiles with those of a model Wilk & Gnanadesikan (1968)
  - sort the data: $r_{(1)} \leq r_{(2)} \leq \cdots \leq r_{(n)}$
  - plot $r_{(i)}$ against the model quantile $F^{-1}\big((i-0.5)/n\big)$
  - $r_{(i)}$: the i-th smallest return; F^{-1}: the quantile function of the model; $(i-0.5)/n$: the share of the data below $r_{(i)}$
- Reading
  - points on the 45-degree line: the model fits
  - an S-shape, low end below and high end above the line: heavier tails than the model
  - only one end bent away: skewness
- Unlike JB, it shows **where** the model fails: centre, shoulders or tails

QQ Plots against the Normal Distribution



- Standardised returns against $N(0,1)$ quantiles: all three bend away from the line at both ends
- S&P 500: lowest standardised return -11.8 where the Normal quantile is -3.7; BET: -11.2 vs -3.7
- The lower tail bends more than the upper one: negative skewness

SFM_ch2_qq_plots

Recap: Testing Normality

- JB combines skewness and kurtosis; it rejects normality for every series
- Report the size of the departure; Normal-theory standard errors are too small
- QQ plots show where the model fails: here, in both tails

"Student": William Sealy Gosset

- Chemist and brewer at Guinness, Dublin, working with small samples of barley and hops
- The brewery did not allow its staff to publish under their own names: he signed "Student" Student (1908)
- He derived the distribution of $(\bar{X} - \mu)/(s/\sqrt{n})$ for Normal data: the t distribution
- In finance it is used for a different reason: its tails are heavier than those of the Normal distribution Praetz (1972), Blattberg & Gonedes (1974)



William Sealy Gosset (1876–1937)

Photo: unknown author (1908); public domain; Wikimedia Commons

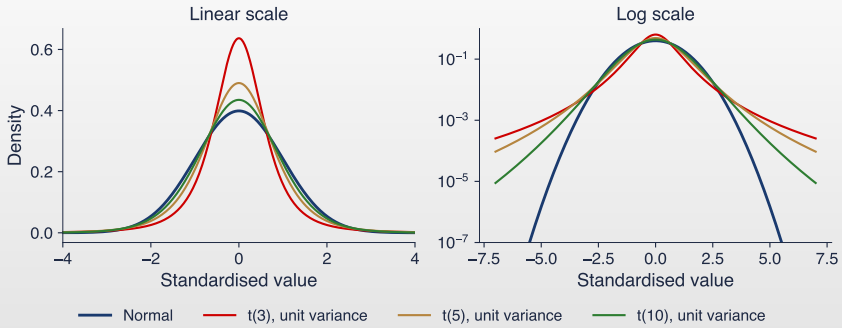
The Student-t Distribution: Definition

- $T = Z/\sqrt{V/\nu}$ with $Z \sim N(0,1)$ independent of $V \sim \chi^2(\nu)$: $T \sim t(\nu)$
  - $\nu > 0$: the **degrees of freedom**, which control the tails
  - $\chi^2(\nu)$: the distribution of a sum of ν squared independent $N(0,1)$ variables; Γ: the gamma function, $\Gamma(k) = (k-1)!$ for integer k
- Density: $f(x) = \dfrac{\Gamma\big((\nu+1)/2\big)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)}\Big(1 + \dfrac{x^2}{\nu}\Big)^{-(\nu+1)/2}$
- Location–scale version for returns: $r = m + s\,T$, with location m, scale s and ν
- As $\nu \to \infty$, $t(\nu) \to N(0,1)$: the Normal distribution is a limiting case

Moments and Tails

- Mean 0 if $\nu > 1$; variance $\nu/(\nu-2)$ if $\nu > 2$; excess kurtosis $6/(\nu-4)$ if $\nu > 4$
- Moments of order ν and higher are infinite
  - $\nu \leq 4$: infinite kurtosis; $\nu \leq 2$: infinite variance
- **Power-law tails**: $P(|T| > x) \approx c\, x^{-\nu}$ for large x
  - $c > 0$: a constant; doubling x divides the tail probability by about 2^{ν}
  - the Normal tail falls like $e^{-x^2/2}$, much faster than any power
  - ν is also the **tail index**, estimated directly in Chapter 5
- To compare with the Normal distribution at the same variance, use $T\sqrt{(\nu-2)/\nu}$, which has variance 1

Student-t Densities with Unit Variance



- Smaller ν: a taller centre and heavier tails; on the log scale the Normal tails drop like a parabola
- $P(|X| > 4)$ at unit variance: $t(3)$ 0.62%, $t(5)$ 0.36%, $t(10)$ 0.12%, Normal distribution 0.0063%
- With $\nu = 5$, a 4-sigma day is 56 times more likely than under the Normal distribution

SFM_ch2_student_t_fit

Fitting a Student-t by Maximum Likelihood

- **MLE** (maximum likelihood estimation): choose (ν, m, s) that maximise the log-likelihood
  - $\ell(\nu, m, s) = \sum_{t=1}^{n} \ln f\big((r_t - m)/s; \nu\big) - n \ln s$
  - $f(\cdot; \nu)$: the $t(\nu)$ density; $-n \ln s$ corrects for rescaling the returns by s
  - no closed form: a numerical optimiser (`scipy.stats.t.fit` in the notebook)
- **AIC** (Akaike information criterion) Akaike (1974): $\text{AIC} = 2k - 2\ell$
  - k = number of parameters (2 for the Normal distribution, 3 for the t); the lower AIC wins
  - $\Delta\text{AIC} = \text{AIC}_{\text{Normal}} - \text{AIC}_t > 0$ favours the t
- Early evidence for stock prices: Praetz (1972), Blattberg & Gonedes (1974)

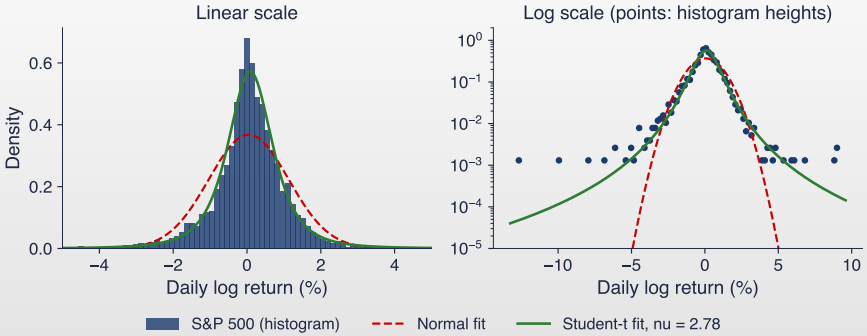
Fitted Student-t: Degrees of Freedom, 2010–2026

Series	$\hat{\nu}$	ΔAIC
S&P 500	2.78	1213
DAX	3.31	739
BET	2.96	1454
Bitcoin	2.23	1335
OMV Petrom (SNP)	3.79	679
BRD	4.67	478
Nuclearelectrica (SNN)	3.08	541
Romgaz (SNG)	3.40	482

- All $\hat{\nu}$ between 2.23 (Bitcoin) and 4.67 (BRD)
- $\hat{\nu} < 4$ for 7 of the 8 series: the fitted model has infinite kurtosis
- ΔAIC is at least 478: the t beats the Normal distribution everywhere

SFM_ch2_student_t_fit

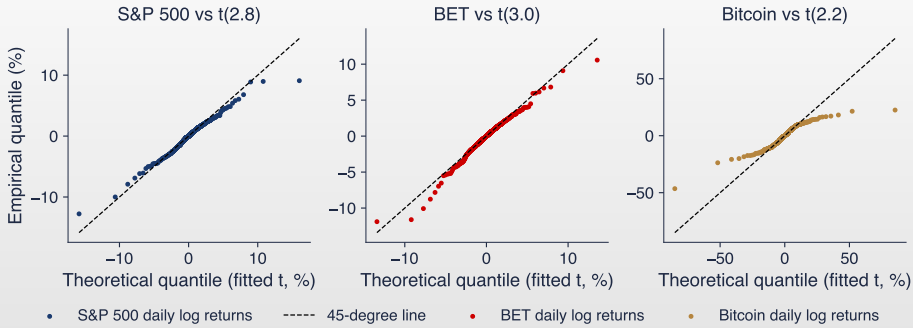
S&P 500: Normal vs Student-t Fit



- Linear scale: the t ($\hat{\nu} = 2.78$) follows the tall centre that the Normal distribution misses
- Log scale: the Normal density collapses beyond $\pm 4\%$; the t follows the observed tails

SFM_ch2_student_t_fit

QQ Plots against the Fitted Student-t



- The centre and the moderate tails now lie on the line
- Extremes: the fitted t expects lower minima than observed: S&P 500 -15.8% vs -12.8%; Bitcoin -85.0% vs -46.5%
- With $\hat{\nu}$ close to 2 the model overshoots the most extreme quantiles

SFM_ch2_qq_plots

What the Student-t Still Misses

- **Symmetry**: the t cannot reproduce the negative skewness
  - skewed t distributions exist; they are beyond this chapter
- **Independence**: an i.i.d. t model has no volatility clustering
  - GARCH models with t errors combine both Bollerslev (1987) (Chapter 9)
- **Single tail index**: one ν for both tails and for the centre
  - extreme value theory models each tail separately (Chapter 5)
- Still the simplest heavy-tailed model, and a large improvement on the Normal distribution

Recap: Student-t

- One extra parameter ν controls the tails; power-law tails, the Normal distribution as $\nu \to \infty$
- Fitted $\hat{\nu}$ between 2 and 5 for daily returns; AIC strongly prefers the t
- It misses skewness and volatility clustering

What Is a Stylised Fact?

- A **stylised fact**: a statistical property shared by many assets, markets and periods Cont (2001)
- Qualitative, not exact: "heavy tails", not "$\nu = 3.2$"
- Rama Cont lists eleven; we check six on our data
- A good model of returns must reproduce them; the benchmark i.i.d. Normal model fails most of them
- Earlier evidence: Mandelbrot (1963), Fama (1965)



Rama Cont

Photo: Renate Schmid (2012); CC BY-SA 2.0 DE; Wikimedia Commons

Six Stylised Facts

- **1. Heavy tails**: large moves are far more frequent than under the Normal distribution
- **2. Aggregational Gaussianity**: returns over longer horizons are closer to the Normal distribution
- **3. Absence of linear autocorrelation**: past returns barely predict future returns
- **4. Volatility clustering**: large moves follow large moves, of either sign
- **5. Leverage effect**: falls raise future volatility more than rises
- **6. Gain/loss asymmetry**: large losses are larger and more frequent than large gains

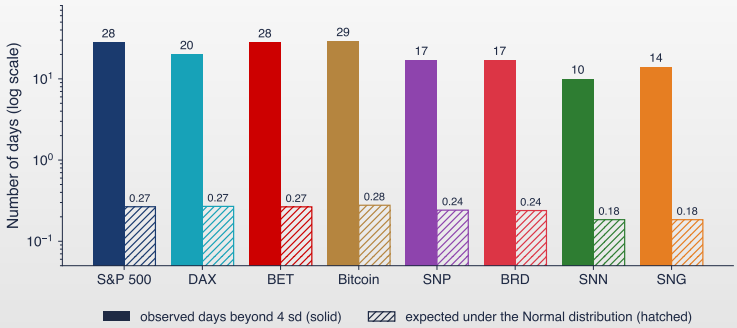
Fact 1: Heavy Tails, Counted

Series	n	$> 3\sigma$	$> 4\sigma$	$> 5\sigma$	$> 6\sigma$	p-value (Poisson, 4σ)
S&P 500	4 201	62	28	12	8	2×10^{-46}
DAX	4 244	56	20	5	4	1×10^{-30}
BET	4 186	70	28	16	11	2×10^{-46}
Bitcoin	4 384	80	29	10	4	6×10^{-48}
Normal distribution, $n = 4\,201$		11.34	0.27	2.4×10^{-3}	8.3×10^{-6}	

- $> k\sigma$: number of days with $|r_t - \bar{r}| > k\,s$; last row: number expected under the Normal distribution
- p-value: probability of at least that many 4-sigma days if the Normal count is Poisson with mean $n \times 6.3 \times 10^{-5}$
- S&P 500: 105 times more 4-sigma days than the Normal distribution allows

SFM_ch2_sigma_days

Fact 1: 4-Sigma Days in Every Market



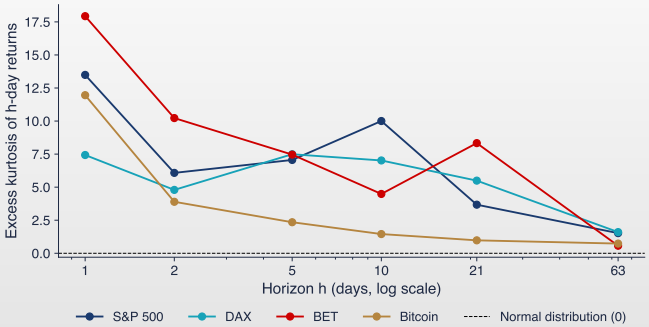
- Observed days beyond 4 standard deviations (log scale) against the Normal expectation of 0.18–0.28 days
- Indices and Bitcoin: one 4-sigma day every 150–212 days; BVB stocks: 10–17 days in total
- Under a fitted Student-t, the expected numbers are close to the observed ones: S&P 500 34.9, BET 28.1 (Seminar 2, B4)

SFM_ch2_sigma_days

Fact 2: Aggregational Gaussianity

- h-day log return: $r_t(h) = r_t + r_{t-1} + \cdots + r_{t-h+1}$, taken on non-overlapping blocks
  - if the CLT applies, the distribution of $r_t(h)$ approaches the Normal distribution as h grows
- The stylised fact: the shape changes with the horizon; it is not the same at all time scales Cont (2001)
- Price to pay: fewer observations
  - S&P 500, 17 years: 4 201 daily, 200 monthly and 66 quarterly returns
  - estimates of kurtosis at long horizons are very noisy

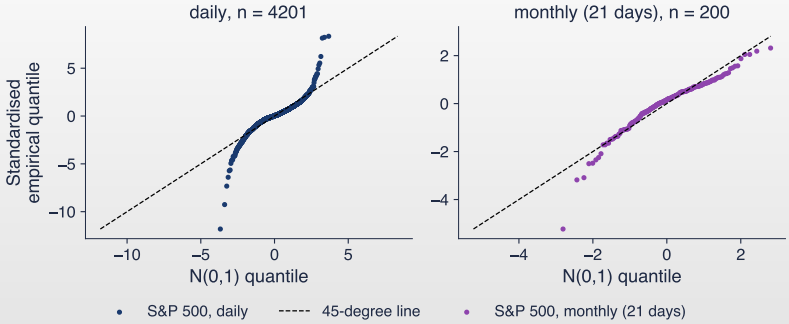
Fact 2: Kurtosis Falls with the Horizon



- Excess kurtosis, daily → monthly → quarterly: S&P 500 13.5 → 3.7 → 1.5; BET 17.9 → 8.3 → 0.6; Bitcoin 12.0 → 1.0 → 0.7
- The fall is slow and not monotone: one crash month (March 2020) dominates a sample of 200 months
- Quarterly returns: JB no longer rejects normality for the BET (p-value 0.07) and Bitcoin (p-value 0.22)

SFM_ch2_cont_stylised_facts

Fact 2: S&P 500, Daily vs Monthly QQ Plots



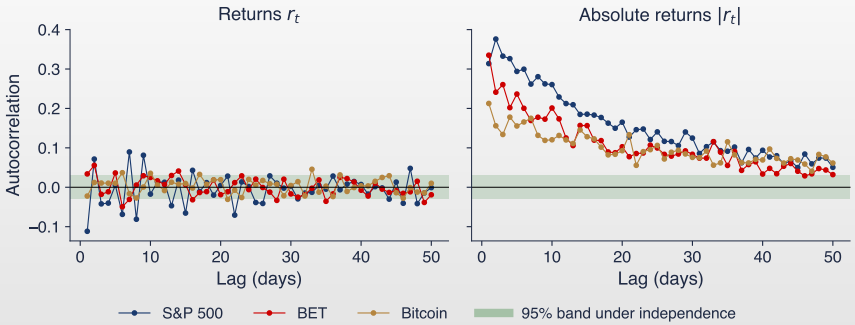
- Monthly returns lie much closer to the line than daily returns
- But monthly returns are not Normal: skewness -1.18, excess kurtosis 3.7, $n = 200$; the lowest point is March 2020
- Why so slow? Volatility clustering makes daily returns dependent, which slows the CLT

SFM_ch2_cont_stylised_facts

Fact 3: Absence of Linear Autocorrelation

- **ACF** (autocorrelation function): $\rho(h) = \text{Corr}(r_t, r_{t-h})$, $h = 1, 2, \ldots$
  - sample version $\hat{\rho}(h) = \sum_t (r_t - \bar{r})(r_{t-h} - \bar{r}) / \sum_t (r_t - \bar{r})^2$
  - h: the lag in days; $\rho(h) \in [-1, 1]$; for i.i.d. data, $\hat{\rho}(h) \approx N(0, 1/n)$: 95% band $\pm 1.96/\sqrt{n}$
- **LB** (Ljung–Box) test Ljung & Box (1978) of H_0: $\rho(1) = \cdots = \rho(m) = 0$
  - $Q(m) = n(n+2) \sum_{h=1}^{m} \hat{\rho}(h)^2/(n-h) \sim \chi^2(m)$; for $m = 10$, reject if $Q > 18.3$
  - m: the number of lags tested together; Q adds up the squared autocorrelations, so large values mean some $\rho(h) \neq 0$
- Why absent? If returns were predictable, traders would exploit it until the predictability disappears (efficient markets, Chapter 7)

Facts 3 and 4: ACF of Returns and of Absolute Returns



- Left: returns; most autocorrelations are inside or close to the band; S&P 500 $\hat{\rho}(1) = -0.11$, BET 0.03, Bitcoin -0.02
- Right: absolute returns; all positive, slowly decaying: S&P 500 0.31 at lag 1, 0.26 at lag 10, 0.05 at lag 50

SFM_ch2_cont_stylised_facts

Fact 3: Small but Not Always Zero

- Ljung–Box $Q(10)$ for returns: S&P 500 199, BET 46, Bitcoin 20; critical value 18.3
  - formally, some autocorrelation is significant
- Two reasons not to over-read this
  - the values are economically small: $|\hat{\rho}(h)| \leq 0.11$, so at most 1.2% of the variance is explained
  - the band $\pm 1.96/\sqrt{n}$ assumes i.i.d. returns; with volatility clustering it is too narrow, so "significant" is found too often
- The S&P 500 value at lag 1 comes mostly from March 2020, when large falls and rebounds alternated: without 20 February–30 April 2020, $\hat{\rho}(1) = -0.04$
- Robust tests of predictability (variance ratios): Chapter 7

Fact 4: Volatility Clustering

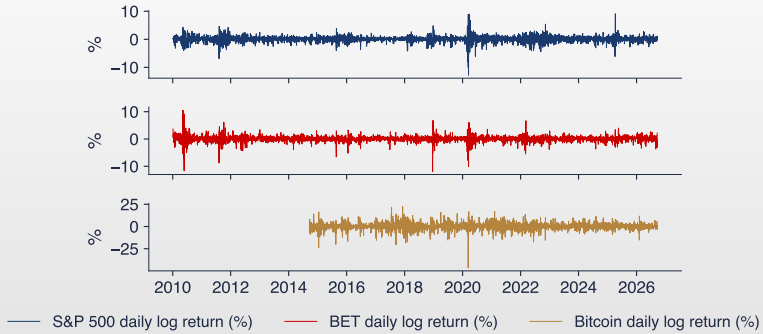
- "[L]arge changes tend to be followed by large changes—of either sign—and small changes tend to be followed by small changes" Mandelbrot (1963)
- The sign of tomorrow's return is unpredictable; its size is not
- Evidence: ACF of $|r_t|$ positive for 50 lags
  - Ljung–Box $Q(10)$ of $|r_t|$: S&P 500 3859, BET 2120, Bitcoin 1086
- So returns are uncorrelated but **not independent**
- Models: ARCH and GARCH (Chapter 9)



Benoit Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 fr; Wikimedia Commons

Fact 4: Calm and Turbulent Periods



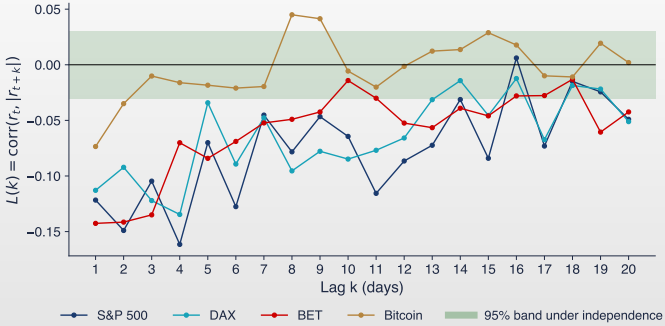
- Turbulent periods: 2010–2011 (euro area debt crisis), December 2018 for the BET, March 2020 everywhere, April 2025
- Calm years in between: the volatility of a single day depends on the period it falls in

SFM_ch2_cont_stylised_facts

Fact 5: The Leverage Effect

- Measure: $L(k) = \text{Corr}(r_t, |r_{t+k}|)$, $k = 1, 2, \ldots$
  - $L(k) < 0$: a fall today is followed by larger moves than a rise of the same size
- Explanation by financial leverage Christie (1982)
  - when the share price falls, the debt-to-equity ratio rises, so equity becomes riskier
- A second explanation, volatility feedback: higher expected volatility raises required returns and lowers prices
- Models with an asymmetric response (EGARCH, GJR-GARCH): Chapter 9

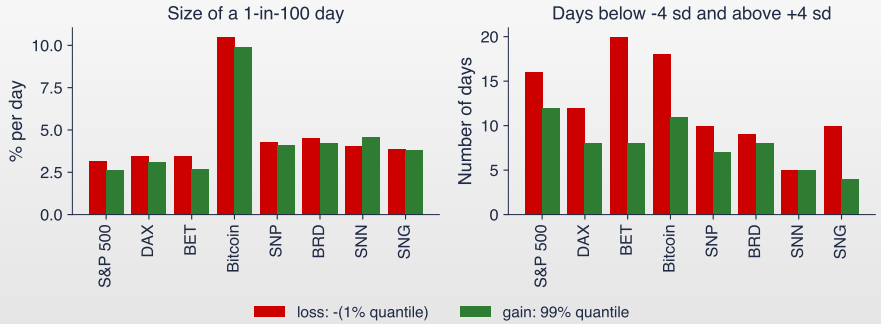
Fact 5: Leverage Correlations



- Stock indices: $L(k) < 0$ for most of the first 20 lags; mean of $L(1..5)$: S&P 500 -0.12, DAX -0.10, BET -0.11
- Bitcoin: $L(1) = -0.07$, then inside the band: no firm has debt behind Bitcoin, so the leverage mechanism is absent

SFM_ch2_cont_stylised_facts

Fact 6: Gain/Loss Asymmetry



- Left: a 1-in-100 loss is larger than a 1-in-100 gain for the indices: BET 3.45% vs 2.66%, S&P 500 3.15% vs 2.61%
- Right: days below -4σ vs above $+4\sigma$: BET 20 vs 8, S&P 500 16 vs 12, Bitcoin 18 vs 11
- Exception: SNN, with positive skewness and a larger right quantile

SFM_ch2_sigma_days

Six Facts, Eight Series

Series	Exc. kurt.	Exc. kurt., 21 days	$\hat{\rho}_r(1)$	$\hat{\rho}_{\lvert r\rvert}(10)$	$\bar{L}(1..5)$	$\lvert q_{0.01}\rvert / q_{0.99}$
S&P 500	13.5	3.7	−0.11	0.26	−0.12	1.21
DAX	7.4	5.5	0.01	0.16	−0.10	1.12
BET	17.9	8.3	0.03	0.20	−0.11	1.30
Bitcoin	12.0	1.0	−0.02	0.12	−0.03	1.07
OMV Petrom (SNP)	8.9	3.3	−0.02	0.13	−0.03	1.04
BRD	7.3	0.4	0.03	0.11	−0.05	1.07
Nuclearelectrica (SNN)	5.1	2.3	0.09	0.08	−0.03	0.88
Romgaz (SNG)	5.3	1.4	0.07	0.08	−0.04	1.02

- Columns: excess kurtosis of daily and 21-day returns; ACF of returns at lag 1 and of $|r_t|$ at lag 10; mean $L(k)$ over lags 1–5; 1% over 99% quantile
- Fact 1 holds for every series, fact 2 with noise; fact 3 only approximately, $|\hat{\rho}_r(1)|$ up to 0.11 (S&P 500; SNN 0.09); fact 4 everywhere, weakest for SNN and SNG (0.08)
- Fact 5: clear for the indices (-0.12 to -0.10), equally weak for Bitcoin, SNP, SNN and SNG (-0.04 to -0.03); fact 6 fails for SNN (0.88), close to 1 for SNP and SNG
- A comparison of the return distributions of cryptocurrencies and traditional assets: Pele et al. (2023)

SFM_ch2_cont_stylised_facts

What the Facts Mean for Models

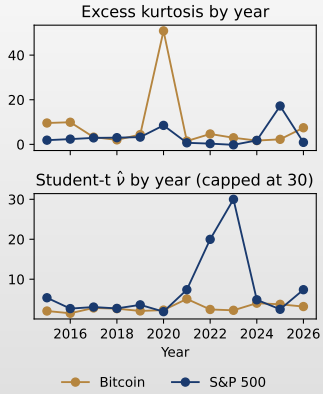
Model	1	2	3	4	5	6
i.i.d. Normal	–	trivially	✓	–	–	–
i.i.d. Student-t	✓	✓	✓	–	–	–
GARCH with t errors (Ch. 9)	✓	✓	✓	✓	–	–
EGARCH, GJR-GARCH (Ch. 9)	✓	✓	✓	✓	✓	(✓)

- Columns: the six facts in the order of this section; (✓): only for returns over several days, unless the errors are skewed
- Risk measures (VaR 1%, ES 2.5%, Chapter 10) inherit the quality of the model: a Normal VaR is too low in crises
- A related summary of tail risk, information entropy: Pele, Lazar & Dufour (2017)

Recap: Stylised Facts

- Heavy tails: 4-sigma days every few months instead of once in decades
- Returns are nearly uncorrelated, but their size is strongly autocorrelated
- Longer horizons are closer to the Normal distribution, slowly
- Stock indices: falls raise volatility, and losses are larger than gains

An Open Question: Are Bitcoin's Tails Getting Thinner?



- As a market matures (more traders, futures, ETFs), its tails might thin out
- Bitcoin, 2015–2026: yearly $\hat{\nu}$ between 1.5 and 5.1; rank correlation with the year 0.63 (p-value 0.03)
- Why it is open: 12 noisy yearly estimates, one year (2020, excess kurtosis 51) dominated by one day
- AI tools can speed up such a study; they do not replace checking it Wang et al. (2023)

SFM_ch2_tails_by_year

How AI Could Help

- **Literature**: find studies of tail indices of crypto assets and summarise their methods
- **Code**: draft rolling-window estimates of ν and of the excess kurtosis
- **Robustness**: propose other windows, other tail measures (Chapter 5), other crypto assets
- Example prompt
  - `Write a Python function that fits a Student-t distribution to daily Bitcoin log returns on rolling 365-day windows and returns the degrees of freedom with a bootstrap 95% interval.`

What to Check

- Data: the same price column and calendar for every year; errors among the largest moves
- Estimation: did the optimiser converge? $\hat{\nu}$ near 2 or above 30 needs a second look
- Inference: one $\hat{\nu}$ per year is noisy; report intervals, not only point estimates
- Volatility clustering: thinner yearly tails may only mean calmer years (Chapter 9)
- References: every cited paper must exist; check the DOI

Project Seed

- **Question**: do the tails of crypto assets thin out as their markets mature?
  - data: Bitcoin, Ethereum and Solana from the course data; S&P 500 as a benchmark
- Steps
  - estimate $\hat{\nu}$ and the excess kurtosis on rolling one-year windows, with bootstrap intervals
  - test for a trend; repeat on returns divided by a rolling volatility
  - compare the count of 4-sigma days per year with the S&P 500
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected

Key Formulas

Quantity	Formula
Normal PDF	$f(x) = (\sigma\sqrt{2\pi})^{-1} \exp\big(-(x-\mu)^2/(2\sigma^2)\big)$
Lognormal moments	median e^m, mean $e^{m+s^2/2}$
CLT	$\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0,1)$
Skewness, excess kurtosis	$S = m_3/m_2^{3/2}, \quad K = m_4/m_2^2 - 3$
Jarque–Bera	$\mathrm{JB} = \frac{n}{6}(S^2 + K^2/4) \sim \chi^2(2)$
Student-$t(\nu)$	variance $\nu/(\nu-2)$, excess kurtosis $6/(\nu-4)$
AIC	$2k - 2\ell$
ACF, Ljung–Box	$\hat{\rho}(h), \quad Q(m) = n(n+2)\sum_{h=1}^{m} \hat{\rho}(h)^2/(n-h)$
Leverage	$L(k) = \mathrm{Corr}(r_t, \lvert r_{t+k}\rvert)$

Check Yourself

- **Question**: daily returns are $N(0.05\%, 1\%^2)$; how many standard deviations is a -3% day?
  - **Answer**: $z = (-3 - 0.05)/1 = -3.05$
- **Question**: $n = 2500$, $S = -0.5$, $K = 6$; what is JB?
  - **Answer**: $\frac{2500}{6}(0.25 + 9) = 3854$; normality is rejected
- **Question**: an annual log return is $N(5\%, 20\%^2)$; what are the median and the mean of 100 invested after one year?
  - **Answer**: median $100e^{0.05} = 105.1$, mean $100e^{0.05+0.02} = 107.3$
- Next: Chapter 3, α-stable distributions

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